Hybrid Atomistic-Parametric Decoherence Model for Molecular Spin Qubits

Katy Aruachan,[1] Sanoj Raj,[2] Yamil J. Colón,[3] Daniel Aravena,[4] and Felipe Herrera[1,5,*]

[1] *Department of Physics, Universidad de Santiago de Chile, Av. Victor Jara 3493, Santiago, Chile;*
[2] *Department of Chemistry and Biochemistry, University of Central Arkansas, Conway, Arkansas 72035, United States;*
[3] *Department of Chemical and Biomolecular Engineering, University of Notre Dame, Notre Dame, Indiana 46556, United States;*
[4] *Department of Materials Chemistry, Universidad de Santiago de Chile, Santiago, Chile;*
[5] *Millennium Institute for Research in Optics, Concepción, Chile.*

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Solid-state molecular qubits with open-shell ground states have great potential for addressability, scalability, and tunability, but understanding the fundamental limits of quantum coherence in these systems is challenging due to the complexity of the qubit environment. To address this, we develop a random Hamiltonian approach where the molecular g-tensor fluctuates due to classical lattice motion obtained from molecular dynamics simulations at constant temperature. Atomistic g-tensor fluctuations are used to construct Redfield quantum master equations that predict the relaxation T_1 and dephasing T_2 times of copper porphyrin qubits in a crystalline framework. Atomistic T_1 predictions due to one-phonon spin-lattice interaction overestimate the available experimental data by orders of magnitude. Quantitative agreement with measurements at all magnetic fields is restored by introducing a magnetic field noise model to describe lattice nuclear spins, with field-dependent noise amplitude in the range $\delta B \sim 10\,\mu\mathrm{T} - 1\,\mathrm{mT}$ for the copper porphyrin system. We show that while T_1 scales as $1/B$ experimentally due to a combination of spin-lattice and magnetic noise contributions, T_2 scales strictly as $1/B^2$ due to low-frequency dephasing processes associated with magnetic field noise. Our work demonstrates the potential of dynamical methods for modeling the open quantum system dynamics of molecular spin qubits.

I. INTRODUCTION

Molecular complexes with low-spin electronic ground states are promising platforms for solid-state quantum information processing [1–3]. Similar to other platforms, such as semiconductor defects [4, 5], molecular spin qubits can be optically addressed [6] and coherently coupled to low-dimensional resonators for quantum state manipulation [7, 8]. The spin coherence time of molecular qubits can be longer than other platforms [9] and further extended using dynamical decoupling [10] or by exploiting clock transitions [11]. A key advantage of molecular systems is the possibility of engineering their environment with angstrom-scale precision using a range of chemical design tools [12–15]. However, future improvements in coherence lifetime of molecular spin qubits require better theoretical and experimental understanding of qubit-reservoir interactions than what is currently available [16]

Multi-scale *ab-initio* modeling approaches have been developed and used for computing T_1 (relaxation) and T_2 (decoherence) times of molecular spin qubits [17–25], providing atomistic insights on the coupling of the qubit spin with vibrational modes and nuclear spins in the crystal lattice. These insights can potentially be used for developing chemical design rules that optimize qubit performance. An atomistic description based on the electronic structure of spin qubit materials has in principle higher predictive power than phenomenological theories developed for electron paramagnetic resonance (EPR) spectroscopy [26, 27], but requires accurate evaluation of a large number of coupling constants that describe the interaction of qubit spins with vibrational normal modes of the lattice or nearby atomic nuclear spins. Spin-lattice coupling constants are obtained through finite-difference evaluation of high-order derivatives of the electronic Hamiltonian with respect to atomic displacements along the $3N-6$ normal modes of the lattice structure [19, 28, 29], with atom numbers $N \sim 10^2$, depending on the unit cell size. In addition to the high computational cost, which could be mitigated with machine learning interpolation [30], numerical gradients with respect to normal mode coordinates can introduce errors if the symmetries of the Hamiltonian are not preserved in the finite difference representation [31].

In this work, we develop a hybrid atomistic-parametric methodology for constructing Redfield quantum master equations that predict T_1 and T_2 times of molecular spin qubits embedded in solid-state crystal frameworks. The method builds on a previous stochastic Hamiltonian formulation of the open quantum system problem [32]. In this formulation, the influence of the environment on the qubit Zeeman Hamiltonian is partitioned into stochastic fluctuations of the gyromagnetic tensor $\delta g_{ij}(t)$, due to lattice phonons and molecular vibrations, plus stochastic fluctuations of the local magnetic field at the qubit location $\delta B_i(t)$, due to induced fields produced by other electron and nuclear spins in the crystal structure [33–35] and effective magnetic fields associated with vibrational orbital angular momentum [28]. To construct the bath spectral density $J(\omega)$, which determines the qubit relaxation and decoherence timescales, we

* felipe.herrera.u@usach.cl

obtain $\delta g_{ij}(t)$ from first principles by sampling the electronic effective spin Hamiltonian of the qubit over an ensemble of molecular dynamics trajectories at constant temperature. No numerical derivatives of the Hamiltonian are evaluated. Spectral densities from dynamical quantum-classical dynamics have been used for optical spectroscopy [36, 37]. Here we adapt the approach for solid-state electron spin dynamics for the first time.

The rest of the paper is organized as follows: Section II describes the open quantum system model and the semi-classical approach for constructing the total spectral density. Section III shows results for T_1 and T_2 as functions of magnetic field for copper porphyrin qubits embedded in a crystal framework. Section IV presents conclusions and perspectives for model improvements and applications.

II. OPEN QUANTUM SYSTEM MODEL

We model the relaxation dynamics of a single molecular spin qubit using Haken–Strobl theory [38–40]. The system is composed of an unpaired electronic spin ($S = 1/2$) in an external magnetic field $\mathbf{B} = \sum_j B_j \mathbf{e}_j$. The molecular spin is embedded in a solid-state matrix and couples to lattice phonons and other nuclear spins present in the crystal structure. In Haken-Strobl theory, bath degrees of freedom are treated implicitly via time-dependent fluctuations that they induce on the total system Hamiltonian $\hat{H}(t) = \hat{H}_S + \hat{H}_{SB}(t)$, where $\hat{H}_S$ is time-averaged over bath-fluctuation timescales. $\hat{H}_{SB}(t)$ contains the explicit fluctuation variables. For $S = 1/2$ molecules, $\hat{H}_S$ is given by [41]

$$\hat{H}_S = \frac{1}{2}\mu_B\, g_{ij} B_i \hat{\sigma}_j + \frac{\hbar}{2} A_{ij}\, \hat{\sigma}_i \hat{I}_j, \tag{1}$$

where B_i are external magnetic field components, g_{ij} are elements of the gyromagnetic g-tensor, $\hat{\sigma}_j$ are the Pauli matrices, $\hat{I}_j$ are nuclear spin angular momentum components of the qubit molecule, and A_{ij} are elements of hyperfine tensor.

Bath-induced fluctuations have the form [32]

$$\hat{H}_{SB}(t) = \frac{1}{2}\mu_B \delta g_{ij}(t) B_i \hat{\sigma}_j + \frac{1}{2}\mu_B \delta B_i(t) g_{ij} \hat{\sigma}_j, \tag{2}$$

where $\delta g_{ij}(t)$ arises from lattice motion and vibrations that modifies the qubit response to external magnetic fields and $\delta B_i(t)$ corresponds to random local magnetic fields produced by nearby nuclear spins in the crystal lattice [42] or effective magnetic fields due to vibrational orbital angular momenta [28]. Fluctuations of the hyperfine tensor A_{ij} are neglected to lowest order and $\langle \delta g_{ij}\, \delta B_k \rangle = 0$ at all times.

The qubit coherence and dephasing times, T_1 and T_2, are obtained from the evolution of system density matrix $\hat{\rho}_S$. In the interaction picture with respect to $\hat{H}_S$, the matrix elements $\rho_{ab} = \langle a|\hat{\rho}_S|b\rangle$ in the basis of system eigenstates $|a\rangle$ evolve according to the Redfield quantum master equation [43]

$$\frac{d}{dt}\rho_{ab}(t) = -\sum_{c,d} R_{ab,cd}\rho_{cd}(t). \tag{3}$$

where $R_{ab,cd}$ are elements of the Redfield tensor encoding the relaxation and decoherence dynamics of the spin qubit. In terms of state-to-state transition rates, Redfield tensor elements can be written as [44]

$$R_{ab,cd} = \frac{1}{\hbar^2}\Bigg\{ \delta_{bd} \sum_e \Gamma_{ae,ec}(\omega_{ce}) - \Gamma_{ca,bd}(\omega_{db}) - \Gamma_{db,ac}(\omega_{ca}) + \delta_{ac} \sum_e \Gamma_{be,ed}(\omega_{de}) \Bigg\} \tag{4}$$

where $\Gamma_{ab,cd}(\omega_{dc})$ are decay rate functions evaluated at the system transition frequencies $\omega_{ab} = \omega_a - \omega_b$. The rates are in turn given by

$$\Gamma_{ab,cd}(\omega_{dc}) = \mathrm{Re}\left[\langle a|\hat{\sigma}_j|b\rangle\langle c|\hat{\sigma}_{j'}|d\rangle J_{jj'}(\omega_{dc})\right] \tag{5}$$

where $J_{jj'}(\omega)$ (units of $\mathrm{J}^2\,\mathrm{Hz}^{-1}$) is the bath spectral density representing the strength of the coupling to the environment at a given system transition frequency and temperature. Following Ref. [32], the total spectral density $J_{jj'}(\omega)$ is partitioned as

$$J_{jj'}(\omega) = J^{\delta g}_{jj'}(\omega) + J^{\delta B}_{jj'}(\omega), \tag{6}$$

where $J^{\delta g}_{jj'}(\omega)$ is the spectral density associated with g-tensor fluctuations and $J^{\delta B}_{jj'}(\omega)$ the spectral density due to local magnetic noise.

To obtain T_1 and T_2 at a given magnetic field strength and temperature, the spectral densities $J^{\delta g}_{jj'}$ and $J^{\delta B}_{jj'}$ are obtained as described below, and Redfield tensor elements are constructed from Eqs. (4) and (5) to solve Eq. (3) for a pure non-equilibrium initial state $\hat{\rho}_S(0) = |\psi\rangle\langle\psi|$. The fully polarized spin-up state $|\psi\rangle = |m_s = +1/2\rangle\,|m_I\rangle$ is used to obtain T_1 from the decay dynamics of $\langle \hat{S}_z(t)\rangle$, and the x-polarized state $|\psi\rangle = (1/\sqrt{2})(|m_s = +1/2\rangle + |m_s = -1/2\rangle) \otimes |m_I\rangle$ is used to obtain T_2 from the decay dynamics of $\langle \hat{S}_x(t)\rangle$. Other initial conditions lead to small changes in T_1 and T_2 that do not affect the order magnitude or scaling with field strength and temperature. Different initial nuclear spin states $|m_I\rangle$ also give similar decay timescales.

Fluctuations of the g-tensor due to phonons can be obtained from classical dynamics simulations of the lattice dynamics and classical autocorrelation functions can be defined as (units of Hz^{-1})

$$G^{\delta g}_{ij}(\omega) = \frac{1}{\sqrt{2\pi}} \int_0^\infty dt \langle \delta g_{ij}(t)\delta g_{ij}(0)\rangle e^{\mathrm{i}\omega t}. \tag{7}$$

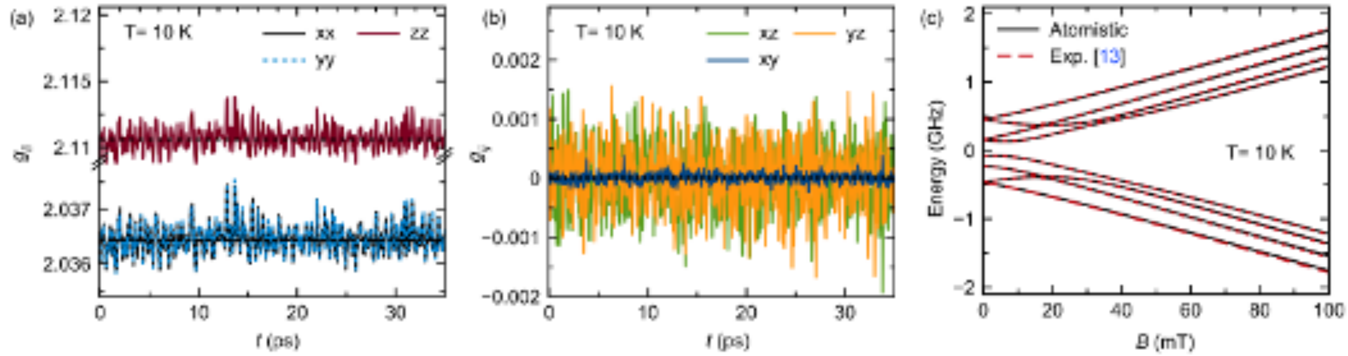

Figure 1. **Atomistic g-tensor fluctuations.** (a) Evolution of the diagonal tensor elements $g_{ii}(t)$ of copper qubits ($S = 1/2, I = 3/2$) in metal-organic frameworks (Cu-PCN-224) at 10 K, sampled from equilibrium molecular dynamics simulation. Time-averaged values are shown in dashed horizontal lines. (b) Off-diagonal elements g_{ij} around their zero-valued time averages at 10 K. (c) Zeeman spectrum of the copper qubit obtained from atomistic time-averaged g-tensor elements at 10K, compared with experimental data from Ref. [13].

Classical bath autocorrelation functions are in general temperature dependent, but do necessarily lead to spectral densities that satisfy detailed balance conditions $\Gamma(\omega)/\Gamma(-\omega) = \exp[\hbar\omega/k_{\rm B}T]$ for Eq. (5). Schemes for constructing quantum spectral densities from classical autocorrelation functions are known [37, 45]. We adopt a harmonic approximation to define the spin-lattice spectral density as

$$J^{\delta g}_{jj'}(\omega) = \delta_{jj'}\,(\mu_B/2)^2\,(\lambda\omega\, B_i^2)\, G^{\delta g}_{ij}(\omega), \tag{8}$$

where λ is an Ohmic parameter set throughout this work as $1/\lambda = 6.9\,\mathrm{cm}^{-1}$. Being second-order in the linear Hamiltonian fluctuation $\delta g(t)$, the spectral density in Eq. (8) models direct one-phonon interaction. Higher-order Raman and Orbach spin-phonon coupling, which tend to dominate the relaxation dynamics of high-S and low-S transition metal solids at temperatures above $T \sim 40-50$ K, [15, 18, 46, 47]), are not included in the model.

Classical autocorrelation functions (ACF) $C_{ij}(t,t') \equiv \delta g_{ij}(t)\delta g_{ij}(t')$ for particular qubit systems can be defined by computing g-tensors for an ensemble of crystal snapshots sampled from equilibrium molecular dynamics (MD) simulations. In this work, we focus on modeling the copper porphyrin qubit system Cu-PCN-224, where the qubit spin is embedded in a metal-organic framework (MOF) crystal. Copper porphyrin transition metal ions Cu^{+2} ($S = 1/2$, $I = 3/2$) are well characterized [48] and in Cu-PCN-224 the copper ion is dilute enough in the lattice to prevent electron spin-spin interactions. EPR spectroscopy and relaxation measurements are available for this qubit system at different magnetic fields and temperatures [13].

MD simulations were performed with Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS) [49] in the NVT ensemble with a time step of 1 fs. The Lammps-interface [50] was used to prepare the input data files using UFF4MOF [51, 52] force field parameters. In our simulation, we made $1\times1\times1$ supercell ($a = b = c = 33.0004$ Å and $\alpha = \beta = \gamma = 109.4712^o$). The supercell has 6 Cu-porphyrin rings. After the simulation, all the rings were extracted for further g-tensor DFT analysis. A non-bonded cutoff of 12.5 Å was applied, with Lennard-Jones potential for the non-bonded interactions and cross-terms were modeled using Lorentz-Berthelot mixing rules. Five simulation replicas were performed at 10 K, 50 K, 100 K, 150 K and 300 K, each simulation running for 1 ns with frames extracted every 50 fs. All DFT calculations were performed using the ORCA 6.0.1 software package [53]. g-factors were calculated from the solution of the response equations to the magnetic field perturbation [54, 55], where the electronic wave function was based on the B3LYP density functional [56] in conjunction with the def2-TZVP basis set [57]. Crystal geometry snapshots taken every 25 steps from the MD simulations were used to compute instantaneous g-tensors. Interpolation was used to join results from different snapshots over a total simulation time of approximately 900 ps.

Local magnetic field noise due to nuclear spins in the lattice is responsible for pure dephasing and low-frequency relaxation of the qubit ($\omega \to 0$) and therefore is not subject to detailed balance conditions. The spin noise spectral density can be written as

$$J^{\delta B}_{jj'}(\omega) = \left(\frac{\mu_B}{2}\right)^2 g_{ij}g_{i'j'}G^{\delta B}_{ii'}(\omega), \tag{9}$$

with magnetic noise spectrum given by (units of $\mathrm{T}^2\,\mathrm{Hz}^{-1}$)

$$G^{\delta B}_{ii'}(\omega) = \frac{1}{\sqrt{2\pi}}\int_0^\infty dt\langle\delta B_i(t)\delta B_{i'}(0)\rangle e^{\mathrm{i}\omega t}. \tag{10}$$

We model the autocorrelation function of the random spin noise as being invariant under rotations (isotropic), of the form

$$\langle \delta B_i(\tau)\delta B_{i'}(0)\rangle \equiv A_B(B)\, e^{-\gamma_{\rm pd}\tau}\delta_{ii'}, \tag{11}$$

with dephasing rate γ_{pd} and field-dependent noise strength A_B given by [32]

$$A_B(B) = a + b\, B^2, \tag{12}$$

where a and b are constants and B is the external magnetic field strength. The magnitude of local magnetic noise is $\delta B \equiv \sqrt{A_B}$. This model is consistent with classical linear response $\delta B \sim \sqrt{b}B$ at moderate fields [58]. The model is parametrized by comparing with available qubit relaxation data.

III. RESULTS AND DISCUSSION

Figure 1a and 1b show the evolution of the diagonal and off-diagonal g-tensor elements $g_{ij}(t)$ of Cu-PCN-224 at $T = 10$ K, computed as described above. By construction, the fluctuations preserve the anisotropy of the g-tensor and the symmetry of the Hamiltonian. By time-averaging the curves over the simulation interval (900 ps), we obtain the stationary elements $\langle g_{xx}\rangle_t = \langle g_{yy}\rangle_t = g_\perp = 2.1106$, $\langle g_{zz}\rangle_t = g_\parallel = 2.0364$ and $\langle g_{xy}\rangle_t = \langle g_{xz}\rangle_t = \langle g_{yz}\rangle_t \ll 10^{-5}$, where $\langle X\rangle_t$ denotes time average. The time-averaged g-tensor is used to construct the Zeeman term in Eq. (1). Hyperfine parameters are not computed from atomistic simulations, but extracted from EPR measurements in Ref. [13] as $A_{zz} = 611$ MHz and $A_{xx} = A_{yy} = 79.4$ MHz. Figure 1c shows the energies of the Zeeman sub-levels obtained by combining the time-averaged atomistic g-tensor with empirical hyperfine parameters. The spectrum agrees with spectroscopic measurements up to minor differences in transition frequencies on the order of 10^{-2} cm^{-1}.

The fluctuations $\delta g_{ij}(t) \equiv g_{ij}(t) - \langle g_{ij}\rangle_t$ are used to construct an ensemble of classical autocorrelation functions $C_{ij}(t,t')$ by segmenting the simulation interval into non-overlapping windows of 35 ps duration each. By averaging over a large number of autocorrelation windows, stationary classical autocorrelations $\langle \delta g_{ij}(\tau)\delta g_{ij}(0)\rangle$ are obtained. Figure 2a shows the spectrum of zz fluctuations $G^{\delta g}_{zz}(\omega)$ obtained from molecular dynamics simulations over a broad range of temperatures $10 - 300$ K. The inset shows a representative autocorrelation function $C_{zz}(t,t')$ over an integration window. The noise level overall increases monotonically with temperature, and frequency noise of the fluctuation data is significant despite the large number of autocorrelation windows used. Similar results are found for other components of the fluctuation tensor $G^{\delta g}_{ij}$.

Figure 2b shows the temperature scaling factor α of the classical fluctuation $G_{ij} \sim T^\alpha$ at different frequencies. On average we obtain $\alpha = 1.03 \pm 0.13$ for G_{xx} and $\alpha = 1.02 \pm 0.14$ for G_{zz} up to 100 cm^{-1}. Other

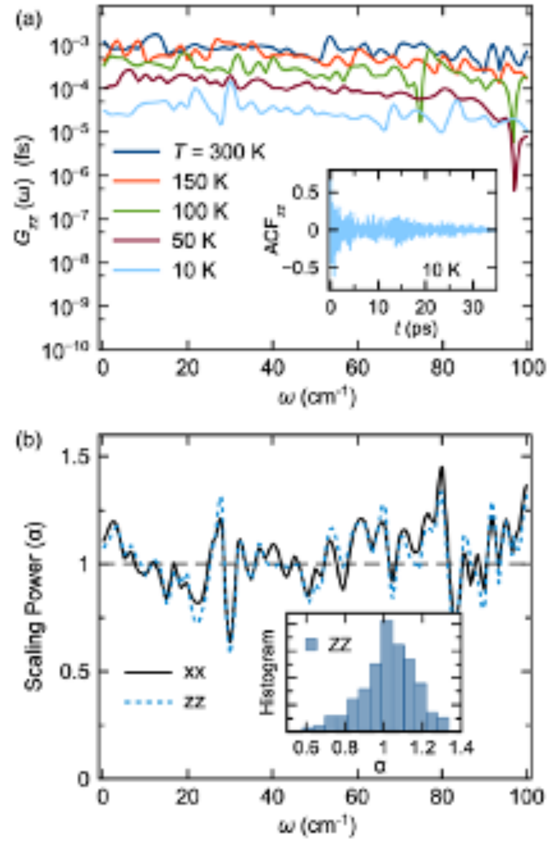

Figure 2. **Spectrum of g-tensor fluctuations.** (a) G_{zz} fluctuation spectrum at different temperatures. The inset shows a sample autocorrelation function (ACF) for zz fluctuations at 10 K. (b) Temperature scaling power α in $G_{ii} \sim T^\alpha$ for xx and zz components, evaluated at different frequencies. The inset shows the histogram of α values for the zz component

spectral components G_{ij} also give $\alpha_{ij} \approx 1.0$. The linear temperature scaling obtained from classical dynamics fluctuations is consistent with direct one-phonon spin-lattice interaction [19, 26]. This scaling is relevant at low temperatures ($<$ 50 K), as direct one-phonon processes do not significantly contribute to T_1 at higher temperatures [15].

Figure 3 shows the total spectral density J_{zz} used to describe the coupling of Cu-PCN-224 qubits with the lattice and spin reservoir at $T = 10$ K. The total spectral density is obtained by combining the atomistic spin-lattice spectral density $J^{\delta g}_{ij}$ with fluctuation spectra in Fig. 2, with a model magnetic field noise spectral density $J^{\delta B}$ given by Eq. (9) with parameters set as $\gamma_{\rm pd} = 0.001$ cm^{-1}, $a = 16\times10^{-10}$ T^2 and $b = 0$. At high frequencies $\omega \gg \gamma_{\rm pd}$, magnetic field noise is suppressed with increasing frequency as $\sim 1/\omega$ due to its Lorentzian character. At low frequencies $\omega \ll \lambda$, coupling to lattice phonons is suppressed due to Ohmic behavior. At intermediate frequencies there is a system-dependent crossover region in which both types of baths contribute

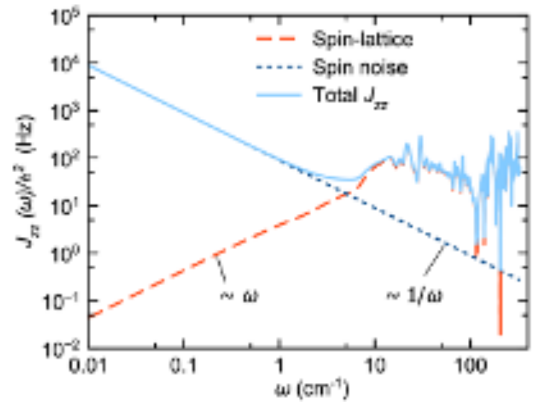

Figure 3. **Hybrid spectral density for copper qubits.** Total spectral density J_{zz} (solid line) obtained by combining the atomistic spectral density $J_{zz}^{\delta g}$ at 10 K (dashed line) with a model magnetic noise spectral density $J_{zz}^{\delta B}$ having field-independent noise amplitude (dotted line; $\gamma_{\rm pd} = 0.001$ cm^{-1}, $a = 16 \times 10^{-10}$ T^2, $b = 0$).

roughly equally to the relaxation dynamics [32]. In our hybrid atomistic-parametric model, at any given temperature the crossover frequency region is controlled by the field noise amplitude parameter a.

Figure 4a shows T_1 predictions for Cu-PCN-224 qubits as a function of magnetic field strength, for different spectral density models. For comparison, experimental measurements from Ref. [13] at 5 K are reproduced ($T = 10$ K data was not available). Results show that taking only the atomistic one-phonon spin-lattice spectral density $J^{\delta g}$ based on the g-tensor fluctuations in Fig. 2a (10 K) overestimates T_1 by at least an order of magnitude at the highest magnetic fields ($B \sim 10\,$T). Since $\omega \sim B$ and $J^{\delta g} \sim \omega B^2$ [Eq. (8)], the atomistic spin-lattice model predicts a $\sim 1/B^3$ divergence of T_1 at vanishing magnetic fields. This zero-field divergence is removed by adding the contribution of magnetic field noise $J^{\delta B}$, which in our model depends on the noise amplitude parameters (a, b) [Eq. (12)]. By assuming a field-independent magnetic noise amplitude ($b = 0$), a crossover field around $B \sim 2.6$ T is established at which T_1 is maximum (Fig. 4a, blue dashed line). The measured T_1 values at low fields are reproduced with noise amplitude $\delta B \equiv \sqrt{a} = 40\,\mu$T. The magnetic field that gives maximum $T_1 \approx 0.72\,$ms in experiments is relatively low (≈ 0.2 T), far below the crossover field predicted by the $b = 0$ noise model. By introducing field-dependence of the noise amplitude with $b = 3 \times 10^{-8}$, the model predictions for T_1 follow closely the experimental measurements across the entire range of magnetic fields (Fig. 4a, red solid line). In this case the field noise amplitude varies from $\delta B \approx 40\,\mu$T at low fields ($10\,$mT) to $\delta B \approx 1.7\,$mT at high fields ($10\,$T). Our spin noise model captures well the measured field scaling $T_1 \sim 1/B$ for Cu-PCN-224 qubits and quantitatively agrees with the six-parameter Brons-van Vleck model, which is commonly used to interpret T_1 vs B measurements (fit

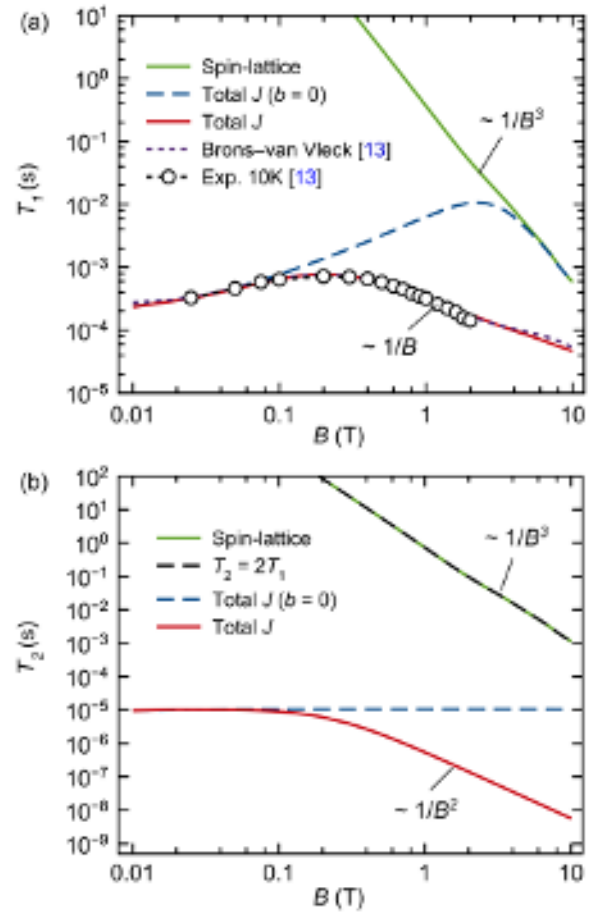

Figure 4. **Relaxation and dephasing of copper porphyrin qubits.** (a) T_1 as a function of field strength B for different spectral density models: atomistic one-phonon spin-lattice relaxation at 10 K (green solid) and hybrid atomistic-parametric spectral density with field-independent (blue dashed) and field-dependent magnetic spin noise (red solid). T_1 measurements and Brons-van Vleck model fit from Ref. [13] are also shown. (b) T_2 as a function of field strength for the models in panel (a). The pure relaxation limit $T_2 = 2T_1$ is shown for comparison (black dashed line). $\gamma_{\rm pd} = 0.001$ cm^{-1} for all magnetic noise models.

parameters taken from Table S18 in Ref. [13]).

Figure 4b shows the dephasing time T_2 as a function of field strength for the same bath models in Fig. 4a. Phase memory times from low-field Hahn-echo measurements of Cu-PCN-224 are in the range $T_m \sim 20 - 50\,$ns up to 80 K [13]. These values roughly set the order of magnitude expected for T_2. As discussed in previous work [32], the spin-lattice spectral density $J^{\delta g}$ gives T_2 values that follow closely the pure relaxation limit $T_2 = 2T_1$ (Fig. 4b, black dashed), because there is no contribution to pure dephasing at $\omega \to 0$ for Ohmic spectral densities. Magnetic field noise gives dephasing contributions that can reduce T_2 relative to T_1 by orders of magnitude. As expected, the magnetic noise model with field-dependent noise amplitude (Fig. 4b, red solid line) gives T_2 values that monotonically decrease with field strength up to $T_2 \sim 10\,\mathrm{ns}$ at the highest fields ($\sim$ 10 T). The field dependence of this model scales as $T_2 \sim 1/B^2$, showing that only near-zero frequency transitions ($\omega \ll \gamma_{\mathrm{pd}}$) are coupled by dephasing channels. Field-independent noise amplitude (Fig. 4b, blue dashed line) gives T_2 roughly constant around $9\,\mu$s. All models predict values of T_2 in the range $\sim 0.001 - 10$ μs depending on field strength, which at low fields tend to be higher than the measured T_m times, although specific pulse sequence modeling should be done for direct comparisons.

IV. CONCLUSIONS AND OUTLOOK

We introduced a general methodology for constructing spectral densities that describe the open quantum system dynamics of copper porphyrin qubits ($S = 1/2$, $I = 3/2$) in a crystalline framework (Cu-PCN-224). The spectral densities are used to construct Redfield quantum master equations that predict relaxation and dephasing times (T_1, T_2) as functions of magnetic field strength. The total bath spectral density $J(\omega)$ is composed of atomistic and phenomenological contributions. The atomistic part describes one-phonon spin-lattice interaction processes through the calculation of g-tensor autocorrelation functions obtained from molecular dynamics simulations and density functional theory. Two-phonon processes are not included in the definition of Redfield tensor, thus limiting the applicability of our atomistic spectral density to cryogenic temperatures [15, 26].

By time-averaging the fluctuations of the g-tensor, the stationary Zeeman spectrum was obtained in good agreement with recent experiments [13]. However, the values of T_1 predicted by the one-phonon spin-lattice spectral density are orders of magnitude longer than experimental observations. We bridge this gap between theory and experiments by introducing a phenomenological spin bath fluctuation model with isotropic noise amplitude $\delta B = \sqrt{a + bB^2}$ [32]. This should be interpreted as an effective description of unresolved long-range electron-electron and electron-spin couplings, through the response of the qubit spin to effective magnetic fields produced by other electron and nuclear spin centers in the crystal structure [35]. Vibrational modes with orbital angular momentum can also induce effective magnetic fields [28]. Components of the spin-induced field noise that are orthogonal to the quantization axis contribute to qubit relaxation, affecting T_1. By comparing with experimental T_1 data for Cu-PCN-224, the magnetic noise amplitude is estimated in the range $\delta B \sim 10\,\mu\mathrm{T} - 1\,\mathrm{mT}$, consistent with first-principles calculations [35] and measurements [11] in other molecular systems.

Fourth-order spin-lattice interaction including Raman and Orbach processes have been shown to dominate the relaxation dynamics for specific molecular systems [20, 28, 59]. These higher order processes would give relaxation rates scaling as the square of the spectral density, $1/T_1 \sim [J(\omega)]^2$ [60], which would imply a magnetic field dependence of T_1 that in principle can also be addressed with the stochastic fluctuation framework proposed here. Given the importance of spin-spin interactions to determine T_1 and T_2 at low temperatures, future method improvements should focus on capturing the influence of nuclear lattice fluctuations on one-phonon and higher-order spin-phonon couplings together with electron-electron and electron-nuclei spin interactions.

Finally, to offer chemical insights for designing of molecular spin qubits with better coherence properties, the structure of the spectral density obtained from time-dependent fluctuations of the qubit Hamiltonian should contain meaningful molecular structure information, which methods based on numerical differentiation of the Hamiltonian along vibrational normal mode coordinates naturally provide [19]. Comparative studies are needed to better assess the strengths and limitations of dynamical fluctuations and static derivatives when modeling decoherence of molecular spin qubits.

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